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Portfolio: github.com/Leo-Marcinkus/Engineering-Portfolio/

Seeking an internship in electrical hardware engineering where I can contribute to hardware design and system reliability through strong troubleshooting skills, while continuing to grow my expertise with hands-on real-world experience.

Projects:

Golden Arduino Board:

Designed and built a custom high-performance Arduino board. Created to streamline future work while completed under a tight deadline. Used Altium, interpreted and implemented multiple datasheets, and analyzed similar microcontrollers to design and assemble a custom Arduino with multiple design improvements.

Retro Glass Fuse Lamp:

Created a retro aesthetic small-scale lamp utilizing glass fuse casings, LED Edison bulbs, and an integrated rechargeable battery circuit to expand my embedded systems and microelectronics knowledge. Improved upon the original inspiration by implementing ESP-NOW control, improvements in power efficiency, and good practices in battery safety resulting in a more robust design.

Alternative Radio Tuning:

Designed a radio receiver utilizing an alternative capacitor-switching LC tuning strategy and a software-controlled frequency selection across the AM/FM broadcast bands. The objective is to find a more power-efficient and higher-performing RF tuning method than the modern method of digital synthesis. Implemented analog switching and audio amplification to receive real-time broadcasts and validate tuning using the NIST 60 kHz reference signal.

Apartment Rover:

Used recycled parts to design a robot to autonomously navigate my apartment to broaden my knowledge of simple circuits. Utilized an Arduino with a proximity sensor and created the speed sensor, direction control, and speed regulation from scratch on a breadboard. Incorporated an OLED display for data visualization. Significantly improved debugging and prototyping skills with reuse constraints.

Sumo Robot:

Developed a basic sumo robot to learn embedded systems. Primarily served to introduce embedded systems and MicroPython. Gained hands-on experience with motor drivers and IR/RF receivers.

Dynacone:

Engineered a dynamically moving sports cone under a tight deadline to gain rapid prototyping experience. It is programmed to execute diverse training patterns and was engineered under a tight deadline with budget constraints. Utilized an Arduino, prototype PCB, motors, and a servo for precise movement.

Lab and Circuit Experience:

Advanced Electronic and Semiconductor Laboratory

August 2025 – December 2025

Analyzed active and passive semiconductor devices (BJTs, diodes, photodiodes, MOSFETs) using oscilloscopes, DMMs, waveform generators, and various power supplies

Performed I-V measurements, parameter extraction, and curve fitting to evaluate device performance and variability using industry-standard software

Analyzed experimental results and correlated data with theoretical models in formal technical reports

PCB Design and Manufacturing

August 2025 – December 2025

Designed multi-layer PCBs in Altium from schematic design to fabrication-ready layouts

Utilized oscilloscopes to conduct signal integrity, crosstalk, and power-rail noise measurements

Applied PDN and layout best practices through breadboard prototyping and PCB implementation

Embedded Software and Real-Time Operating Systems

August 2025 – April 2026

Developed embedded firmware on ARM Cortex-M microcontrollers using C, registers, and memory-mapped peripherals

Designed multi-threaded real-time applications using FreeRTOS with semaphores, mutexes, and task scheduling

Implemented debugging, tracing, and profiling to validate timing and ensure optimal system performance

Electronics Design Lab

August 2024 – December 2024

Designed, tested, and modeled DC motors, active op-amp low-pass filters, and encoder interface circuits for optimal signal conditioning

Performed DC, transient, and frequency domain analysis in the SPICE-based software SIMetrix to optimize and debug circuit performance

Skills:

Hardware and PCB Design

Design boards that pass validation quicker and with reduced debugging time

Saturn PCB, multi-layer layouts, schematic testing, Altium, signal integrity, PDN design, power management

Electronics Analysis

Perform circuit debugging and signal integrity analysis

Analog circuit design, circuit testing, I-V characterization, LTSpice, ADS, HyperLynx

Embedded Systems

Develop reliable firmware for rapid prototyping under firm deadlines

STM32 boards, Arduino, microcontrollers, PWM, SPI/I²C/UART, Real-Time Systems, RISC-V

Programming Tools and Systems

Analyze experimental data and automate data collection

C/C++, Python/MicroPython, SystemVerilog, Assembly, CAD

Testing Tools and Verification

Rapidly diagnose hardware problems and troubleshoot more effectively

Oscilloscope, logic analyzers, DMM, data validation, waveform generators

Education:

University of Colorado Boulder, Boulder, CO

B.S. [Electrical Engineering](#) - College of Engineering and Applied Science – Expected May 2027

Accelerated M.S. [Electrical Engineering](#) – College of Engineering and Applied Science – Expected May 2028

Interests/Hobbies:

Audio electronics and production, mechanical watches, guitar, piano, NBA, benchmarking components, Jimmy John’s, EDC

Work Experience:

Lanzi Designs and Cabinetry – Independent Contractor

December 2022 – January 2026

Measured, cut, and assembled cabinets with precision and professionally engaged with clients to ensure specifications

Installed cabinetry, flooring, trim, and ceiling support for structural integrity and aesthetics

Eddie Bauer – Sales Associate

September 2022 – May 2023

Operated the register, processed transactions, and assisted customers with purchases

Resolved customer concerns and returns through clear communication and conflict management

Express – Sales Associate

May 2022 – August 2022

Engaged with customers to provide styling advice and case-by-case recommendations

Assisted in closing responsibilities to prepare the store for corporate reviews with executives